

UW ECONOMICS SOCIETY



[HOME](#) [ABOUT US](#) [EXECUTIVE TEAM](#) [ORIGINAL CONTENT](#) [FEATURES](#) [STAY CONNECTED](#)



ORIGINAL ARTICLES May 31, 2023 ► 1 Comment

 Finn
O'Connor

Finn
O'Connor

The Great Leap Sideways – Chinese Industrialization Under Mao Zedong

Written By: Finn O'Connor

After forming the People's Republic of China (PRC) in 1949, Mao attempted to rapidly develop China's economy through a series of centrally administered 5-year plans based on the Stalinist industrial policy. The most famous of these plans is the second one between 1958 and 1962, commonly referred to as the Great Leap Forward (GLF). These plans came at an enormous economic and social cost, resulting in the death of between 15 and 55 million people through one of, if not, the largest famine in history. The failure of Mao's economic policy led him to double-down on the Chinese Communist Party's social control, purging dissidents and rewriting Chinese history through the Cultural Revolution of 1966 to 1976. After the Chairman's death and subsequent reorganization of power within the CCP, Deng Xiaoping undertook broad economic reforms that moved the Chinese economy away from its command structure to a more free-market approach through the household responsibility and Town-Village Enterprise (TVE) programs that stimulated market-driven economic growth. Paired with an opening-up policy that ended decades of economic isolation from the world economy, Deng cultivated some of the most rapid development in history, setting the PRC on track to become one of the strongest of today's economies.

There is no doubt that Deng's reforms revitalized the ailing Chinese economy, but the common history positions Mao's economic strategy opposite to his successors. Mao represents the failures of a strict command system, whereas Deng demonstrates the potential of a free-market system. However, I propose that the Mao and Deng periods of economic policy are more continuous than the commonly understood history suggests. Mao, while orchestrating one of the greatest humanitarian disasters in living memory, also established the foundations for industrialization that Deng and subsequent CCP leaders used to facilitate China's spectacular economic growth. Put rather simply, Deng could not have achieved what he did without Mao's disastrous policies.

Industrialization Defined

This paper draws heavily from the idea of “New Stage Theory” proposed by Yi Wen in his book *The Making of an Economic Superpower*. Wen breaks market-transformation up into a series of steps which must be completed sequentially to achieve industrialization. They are, in Wen’s words:

“Agrarian market structure to proto-industrial market structure and to light-industrial market structure and then to heavy-industrial market structure and finally to service-oriented welfare-state market structure.”(1)

It is this first step, the agrarian to proto-industrial market structure that I will be focusing on.

A term first coined by Franklin Mendels, proto-industrialization refers to the small-scale commercial industry predating the heavy manufacturing that characterizes the traditional notion of an “industrial revolution.” To Wen, it is critical to further development for a multitude of reasons. Including, it improves diversification and labour utilization, trains the workforce in more advanced manufacturing techniques, and increases their purchasing power. On a larger scale, it stimulates infrastructure development through higher government revenues, and facilitates intra-regional trade and exchanges through specialization (2). In essence, proto-industrialization prepares an economy for full-scale industrialization, hence why Wen considers this step indispensable. However, Wen considers Deng’s reforms to be the first real instance of proto-industrialization in the Chinese economy, giving Mao only limited credit, such as mentioning the importance of the communal spirit that Maoist decrees instilled into peasants (3). Yet, while Mao’s policies did not reflect the traditional image of proto-industrialization, many of them (sometimes unintentionally) performed the same functions that make it a critical stepping-stone to true industrialization.

Mao’s Agricultural Policies

It might seem counterintuitive, in an evaluation of Mao's role in Chinese industrialization, to look at his agricultural policy. It seems doubly so given the undisputed failure of the Great Leap Forward as an economic strategy. After all, the Great Leap is an outright rejection of the basic laws of neoclassical economics necessary for economic growth. Heavy-handed centralization removed any incentive for production beyond extractive state quotas, while the arrangement of farmers into self-sustained communes depressed the inter-community trade that could facilitate specialization and division of labour. Indeed, there is a reason why both shortcomings would be addressed by Deng's economic reforms. Yet to grasp the full picture of Mao's Great Leap we must look at it in the context of the Chairman's overarching objective. Mao based his industrial policy on the Soviet strategy, so, like Stalin, he considered agriculture a stepping-stone towards an industrial economy. That is, in the words of Harvard's Adam Bergson, "steel was a final good..., and bread an intermediate one." (4) Mao followed in Stalin's footsteps and prioritized heavy industry, feeding agricultural outputs into urban investments. This is why, as observed by Anthony Tang in 1967, agricultural and industrial growth moved in tandem between 1949 and 1957 (5).



A 1953 Chinese propaganda poster that roughly translates “Study the advanced production experience of the Soviet Union, struggle for the industrialization of our country.” As is the case with much of the PRC's early history, the CCP tried to mirror Soviet methods, often to disastrous effect (see Trofim Lyenko)(6).

However, industrial growth built on investments from agriculture constrained industrial development and created bottlenecks when agricultural production dropped, as was often the case during the bouts of bad weather that plagued the Chinese countryside (7). This model is often referred to as the “unbalanced growth” model (as opposed to the “balanced growth” model of the brief 1920s Soviet New Economic Policy), and led to constant cycles of rapid growth, steep slowdowns, and periods of recovery (8). Given the importance of agriculture to the development of the heavy industry, many of Mao's policies that constituted a form of proto-industrialization stemmed from the desire to improve agricultural output to overcome these bottlenecks. Thus, it is important to approach Mao's policies from this industrial-focused perspective as, while the results of the unbalanced growth model failed to utilize China's massive amounts of resources and population, those

policies still produced latent benefits that would only be realized after the Chairman's death.

Diversification

Diversification is one of the most important factors of proto-industrialization. For one, expanding into more industries protects a market from serious shocks and contributes to sustainable growth. Diversification also facilitates further trade and leads to more division of labour, enhancing utilization rates and therefore making the market more efficient.

One element of Mao's Great Leap is the "five small industries" campaign where rural communities would produce key industrial inputs and machinery for other agricultural regions (9). Though this program is better known by its predecessor, the ill-conceived "Backyard Furnaces" campaign that attempted to circumvent the high capital investments required for large-scale factories by establishing smaller, crudely built plants in rural communes. While the backyard furnaces would ultimately fail after only a few months, producing sub-par outputs and pulling labour from much needed agricultural production, the small industries program made use of more modern technology to produce key goods such as steel, fertilizer and even hydropower. These plants were similarly rife with inefficiencies, often resulting in diseconomies of scale due to their small size and amateur workforce, but they also contributed to early diversification as rural markets found other industries beyond agriculture. So much so that the rural industries that sprung up from Deng's TVEs were built on the groundwork of the five industries implemented under Mao (10).



The “Backyard Furnaces campaign” is often up there with the “Four Pests” as the most ridiculed moments of Mao’s reign and demonstrated how a straightforward and well-intentioned mandate could have disastrous knock-on effects (11).

Infrastructure

State-led infrastructure development is the other sector where Wen concedes that Mao played a pivotal role since this is a component of proto-industrialization in which the Chairman had the most direct impact on. Though traditional proto-industrialization generates government revenues through increased economic activity, Mao built up China’s infrastructure using mass mobilization and centrally directed government spending.

For one, the state built sprawling railways during the first five-year plan. Though they would be pushed to overcapacity during the Great Leap Forward and construction of new lines would be reduced by the subsequent economic slowdown (13). But, the rail grid would be utilized to great effect during the reform period as it significantly lowered transport costs between what would be otherwise isolated regions (14).

Unlike rail infrastructure, which scholars generally agree can be at least partially attributed to Mao, his irrigation and drainage projects are widely considered to be ecological and economic disasters. Yet, these

projects had latent benefits that would not actually be noticeable until the post-Mao era of reform. As Y.Y. Kueh points out, irrigation and drainage projects caused massive input cost spikes at the time of development, but we did not see a large enough effect on output to generate positive measures of productivity (13). Kueh argues that we can discern a positive impact on production by comparing the performance of agriculture in the 1930s and the 1950s, where the latter achieved superior productivity over the former, even though both periods utilized the same level of technology. Thus, there is a noticeable improvement as a result of Mao's infrastructure policy (15). However, Chinese agriculture still needed to achieve a level of technology advanced enough to realize the full benefits of Mao's projects, which did not occur until the 1960s. Essentially, as we see even today, it takes time for large infrastructure projects to bear fruit, doubly so in a country like China that was still modernizing. Therefore, the mass mobilization of the Great Leap produced impressive infrastructure that was not utilized during Mao's reign. Yet, the investments that the Chairman made would have a latent effect on economic growth in the 1980s.

Diffusion of Industrial Knowledge

Another key component of proto-industrialization is the development of techniques and knowledge of advanced production methods. From a traditional perspective, this means that labourers refine their skills in textiles and handicrafts which necessitate small-scale firms slightly larger than the cottage industry means of production. This component is one of the largest ways Mao's policies diverged from the traditional notion of proto-industrialization yet achieved the same result. Two programs in the 1960s, the Third Front and the Xiafang campaigns, stand out as an unique, and accidental, means of diffusing industrial knowledge into rural communities.

The Third Front initiative intended to develop industry in the Chinese interior to supplement the vulnerable coast in case of a full-scale invasion of the Chinese mainland. The goal was to provide the nation

with a more strategic depth and complement Mao's doctrine of a People's War, where the PLA would give up urban areas and fight a protracted war of attrition in the countryside. The Third Front, then, would prevent the complete annihilation of Chinese industrial capacity in times of war. But, like many of Mao's economic policies, it is generally considered a failure. Put simply, it was too large a project to be completed at China's level of development. Ambitious development sapped investment from other areas and left projects both in the interior and the coast incomplete (16). However, the development of heavy industry in the interior pulled rural workers into cities to work in factories, giving them the basic expertise that would aid them in rapidly developing these sectors during the late 1970s (17).

On top of the Third Front, Mao also sent young urban-educated workers into rural communities after the GLF through the Shang Shan Xia Xiang (up to the mountains, down to the villages) relocation program. While the intention was to give these privileged urban youth an experience in the countryside to round out their education, the relocation program, as is the case with most forced migration campaigns, came at an enormous social cost. The children selected to be relocated suffered from destroyed social networks and life-long damage to their overall wellbeing (18). Yet, like the Third Front Program, the "Send-Down" diffused industrial knowledge and expertise into rural communities, providing the basis for later economic development. While the generally well educated urban-youth lacked significant factory experience, they brought with them managerial and accounting skills essential for mid-to-large-sized enterprises. They also leveraged their connection to urban centers to import technology from and access markets in large cities, as is what happened between Wuxi County and Shanghai (19).

In summary, Mao's rural-focused policies may seem ill-conceived and poorly implemented, and in many cases they were. They imposed significant economic and social costs on the Chinese people that gave rise to Deng's pivotal reforms. But approached from Mao's perspective,

where agriculture is subservient to industrialization, his policies did ultimately strengthen China's industrial base, even if some of these outcomes were by-products of other goals. Mao's small industries campaign actually contributed to the speed at which rural communities were able to leverage the market opening provided by Deng.

But we leave one critical question unanswered – if Mao's policies checked all the boxes for successful proto-industrialization, then why did it take until the 1980s for China to industrialize? It's easy to say that China simply required access to a global market, but the issue lies deeper within Mao's economic philosophy.

The Barrier to Industrialization

I used Wen's NST framework for this paper as it emphasizes the importance of proto-industrialization as a foundation for further evolution. However, Wen does not simply chart the evolution of production, his steps of market transformation require a shift in domestic consumption that can receive that increased output. Wen defines industrialization as an economy in which there is both the mass production of goods, and a mass market to consume those goods. Recall, Mao's economic policy specifically required suppressing domestic consumption to invest the bulk of outputs into urban centers (20). Critically, the Mao era saw limited foreign trade, restricting the size of the market to Chinese border through a strict "self-reliance" policy. Deng's "opening-up" policy, on the other hand, dramatically increased the volume of trade, giving rising Chinese firms an international market on which to offload output.

Therefore, Mao's policies did not fail to kick-off an industrial revolution because there was no market for those goods beyond the state who controlled the direction of outputs. This was no accident, the Stalinist approach to economic development required the depression of the general population's consumption. The rise of domestic consumption,

as well as the growing availability of foreign markets under Deng, fueled China's budding manufacturing industry.

So, as we can see, Mao's policies not only provided a basis for rapid growth in heavy industry, but had knock-on effects that set the foundations for a broad market transformation. As such, we can confidently conclude that proto-industrialization occurred prior to Deng's reforms. Mao's agricultural policies developed new industries in the people's communes and developed irrigation systems that provided the basis for rural development in the 1980s. He also inadvertently diffused essential industrial knowledge into rural communities through the Third Front and Send-down programs. However, due Mao's Soviet-style "unbalanced" development model, general consumption was depressed in favour of state-directed investment. That is, the Chairman failed to cultivate the mass-market required by mass-production, freezing China in the proto-industrial state of development.

Conclusion

The big lesson here is that China's industrialization process is unique and significantly different from Western countries. Mao's economic policies were not tailored to establishing the foundations for future development but implemented to build-up the crown jewel of the post-1949 economy, a powerful heavy industry. Thus, while this paper focuses on the critical proto-industrialization under Mao, this is not to ignore developments in other sectors such as a thriving light-industry sector or comprehensive welfare systems. An often-overlooked policy under Mao was the decentralization of regional industries in 1956-1958 that facilitated market-driven exchanges between rural and urban centers of production, often used to great effect by cities like Shanghai, which remains one of the great Chinese commercial powerhouses (21). Perhaps China did not need to follow the path that Wen prescribed but had actually achieved all the components required for industrialization and simply needed one factor, a mass market to consume production,

before they could all converge and generate the incredible growth witnessed to this day. Unfortunately, this proposition is beyond the scope of this paper and should be the subject of further research (and perhaps the subject for a future paper). All we can say for certain is that Mao, despite his many missteps, established the proto-industrial foundation required for the rapid growth achieved after his death.

Works Cited

1. Yi Wen, *Making Of An Economic Superpower, The: Unlocking China's Secret Of Rapid Industrialization* (New Jersey: World Scientific Publishing, 2016), 31–32.
2. Wen 399–400
3. Wen, 402–3
4. Anthony M. Tang, “Trend, Policy Cycle, and Weather Disturbance in Chinese Agriculture, 1952–1978,” *American Journal of Agricultural Economics* 62, no. 2 (1980): 339, <https://doi.org/10.2307/1239711>.
5. Anthony M. Tang, “Agriculture in the Industrialization of Communist China and the Soviet Union,” *Journal of Farm Economics* 49, no. 5 (1967): 1127–28, <https://doi.org/10.2307/1236991>.
6. “A Collection: Backyard Furnaces in Henan 1958–1961,” *Everyday Life in Mao's China* (blog), June 8, 2018, <https://everydaylifeinmaoistchina.org/2018/06/08/a-collection-backyard-furnaces-in-henan-1958-1961/>.
7. Tang, 1128.
8. Qunhui Huang, *China's Industrialization Process*, 1st ed. 2018 edition (New York, NY: Springer, 2018), 12.
9. Christine P. W. Wong, “Rural Industrialization in China: Development of the ‘Five Small Industries’” (Ph.D., United States — California, University of California, Berkeley, n.d.), 3, <https://www.proquest.com/docview/302912800/citation/3D7C7C09A3E24B5APQ/1>.
10. Y. Y. Kueh, “Mao and Agriculture in China's Industrialization: Three Antitheses in a 50-Year Perspective,” *The China Quarterly*, no. 187 (2006): 722.

11. Miles Maochun Yu, "The 1969 Sino-Soviet Border Conflicts As A Key Turning Point Of The Cold War," Hoover Institution, n.d., <https://www.hoover.org/research/1969-sino-soviet-border-conflicts-key-turning-point-cold-war>.
12. Rudi Volti, review of *Review of Railroads and the Transformation of China*, by Elisabeth Köll, *China Review International* 24, no. 4 (2017): 312.
13. C. X. George Wei, "Mao's Legacy Revisited: Its Lasting Impact on China and Post-Mao Era Reform," *Asian Politics & Policy* 3, no. 1 (2011): 11, <https://doi.org/10.1111/j.1943-0787.2010.01236.x>.
14. Kueh, "Mao and Agriculture in China's Industrialization," 716.
15. Kueh, 715.
16. Barry Naughton, "The Third Front: Defence Industrialization in the Chinese Interior," *The China Quarterly*, no. 115 (1988): 38.
17. Chris Bramall, "Learning to Industrialize in the Maoist Era," in *The Industrialization of Rural China* (Oxford University Press, 2006), 147, <https://doi.org/10.1093/acprof:oso/9780199275939.003.0006>.
18. Shun Wang and Weina Zhou, "The Unintended Long-Term Consequences of Mao's Mass Send-Down Movement: Marriage, Social Network, and Happiness," *World Development* 90 (February 1, 2017): 15, <https://doi.org/10.1016/j.worlddev.2016.10.011>.
19. Bramall, "Learning to Industrialize in the Maoist Era," 149.
20. *Economic growth and equality* (7)
21. Wei, "Mao's Legacy Revisited," 12.

◆ Banner

Previous Post

Economics Newsletter – May 19,
2023

Next Post

Economics Newsletter – June 2,
2023



CONSULTEZ NOTRE PHARMACIE EN LIGNE À PRIX RÉDUIT DÈS MAINTENANT APRIL 1, 2024 AT 7:52 AM

REPLY ↩

imite sans souci de disponibilité acheter acticin en Espagne en ligne
Commandez votre imite en ligne en toute sécurité
acticin en vente en Italie
où trouver du imite achat acticin en Espagne sur internet
acticin sûre et efficace en ligne
acticin de qualité pharmaceutique en vente
Acheter imite en France sans ordonnance acheter imite en Europe
imite sans complications ni obstacles acticin vente en ligne en Suisse
acheter du imite en Espagne en toute légalité
Où acheter acticin en ligne légalement et en toute sécurité
acticin en pharmacie en ligne : disponibilité et fiabilité
imite en ligne : tout ce que vous devez savoir Comment utiliser acticin en toute sécurité
Commandez votre acticin en ligne France
Commandez votre imite en ligne dès aujourd'hui Acheter du imite de qualité pharmaceutique en ligne
Commandez votre acticin en ligne sans souci
Acheter acticin en ligne en France : conseils pratiques imite disponible sur ordonnance
achat de imite en Europe acticin pas cher : économisez sur votre traitement
achat imite à Marseille acticin en ligne : Ce que les consommateurs doivent savoir
où trouver imite en Suisse acheter acticin sans risque France
acticin disponible sans prescription
Commander imite en ligne où acheter du acticin légalement
Achat de imite sur Internet : ce qu'il faut garder à l'esprit Consultez un professionnel de la santé pour la
imite en ligne
commander acticin France ordonnance médicale

imite disponible en pharmacie suisse

Acheter de la acticin en ligne sans risque en Belgique acticin en ligne : conseils

pour une expérience d'achat sans souci

Commandez acticin en ligne dès maintenant imite sans ordonnance :

Risques et bénéfices

Commandez votre imite en ligne dès aujourd'hui

acticin : comment l'utiliser de manière responsable

Pharmacie en ligne française pour acheter du acticin où acheter du acticin

Achat sécurisé de acticin en ligne en France

acticin à commander sans ordonnance

Acheter de la acticin en ligne : Conseils pratiques acticin :

guide d'utilisation et informations importantes

Achat facile de imite en France

acticin sans prescription : Comment procéder ?

acticin authentique et fiable acticin disponible en ligne avec livraison sécurisée en France

acheter acticin sans problème Belgique

acticin en vente libre en Suisse

Leave a Reply

Your email address will not be published. Required fields are marked *

COMMENT



NAME *

EMAIL *

WEBSITE

POST COMMENT

☐ NOTIFY ME OF FOLLOW-UP COMMENTS BY EMAIL.

☐ NOTIFY ME OF NEW POSTS BY EMAIL.

Copyright 2025 UW Economics Society